Task 5 – Exploratory Data Analysis (EDA)

Dataset: Titanic passenger dataset (train.csv)

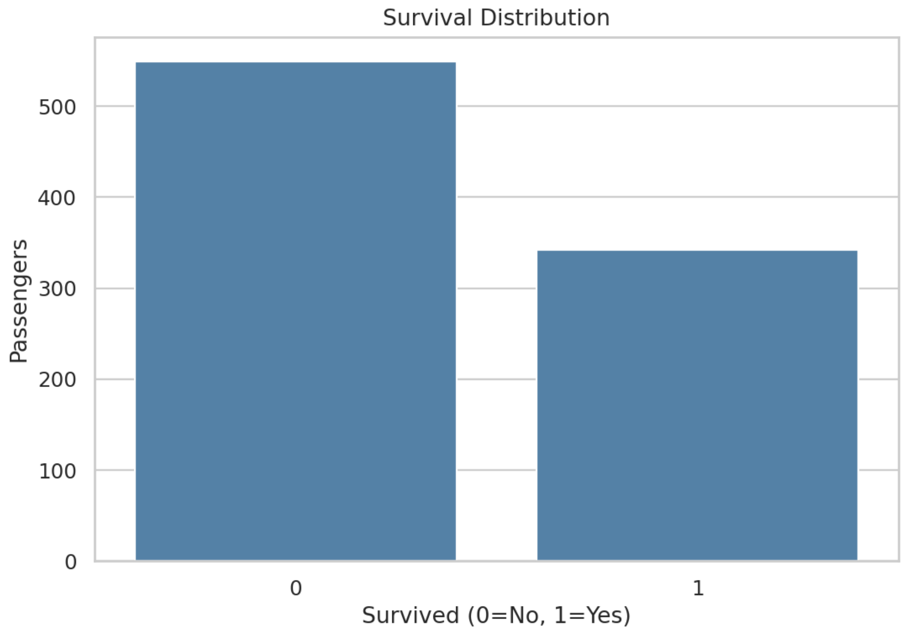
Objective: Extract insights using visual and statistical exploration.

1. Dataset Overview

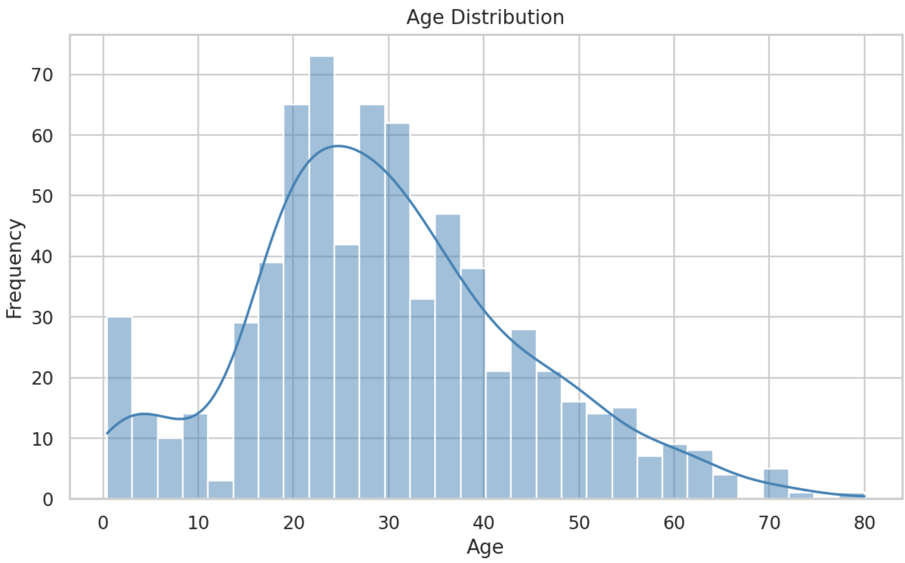
The dataset contains 891 passenger records and 12 columns. The target variable is Survived, where 0 means did not survive and 1 means survived. Age has 177 missing values, Cabin has 687 missing values, and Embarked has 2 missing values.

Metric	Value
Passengers	891
Columns	12
Survived	342 (38.38%)
Did not survive	549 (61.62%)
Missing Age	177 (19.87%)
Missing Cabin	687 (77.10%)
Missing Embarked	2 (0.22%)

2. Univariate Analysis

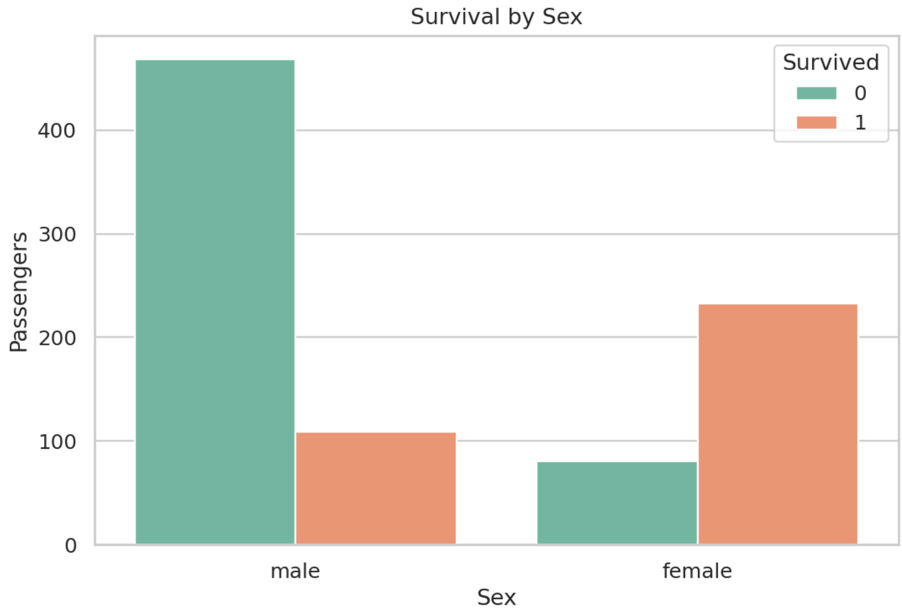


Observation: More passengers did not survive than survived. The overall survival rate is 38.38%.

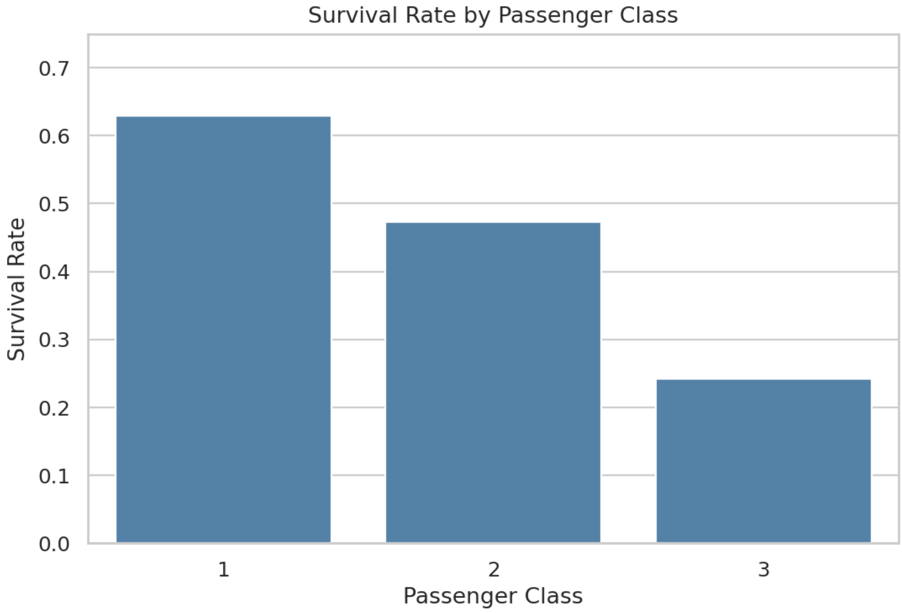


Observation: Age is concentrated around young and middle-aged adults, with fewer very young and elderly passengers.

3. Bivariate Analysis

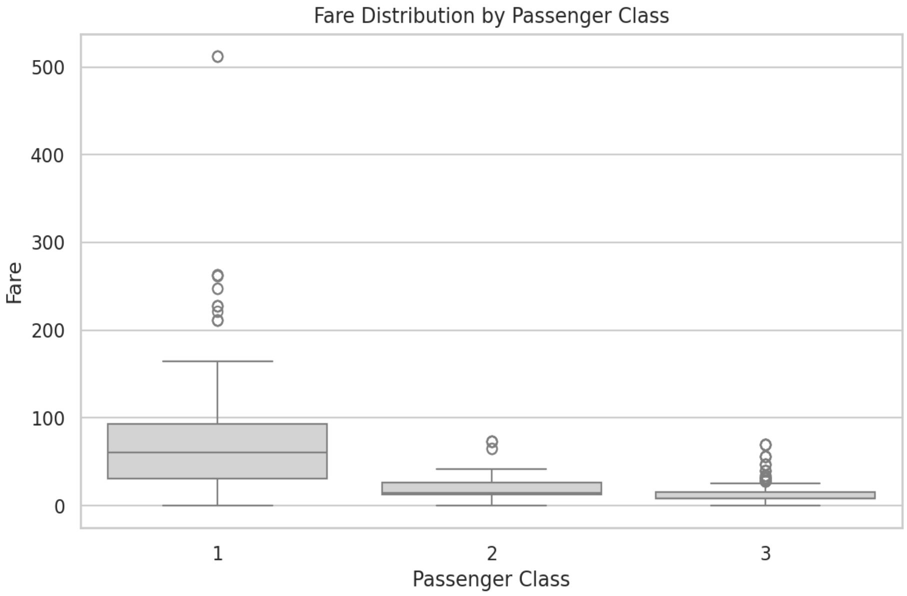


Observation: Female survival was about 74.2%, compared with about 18.9% for males.

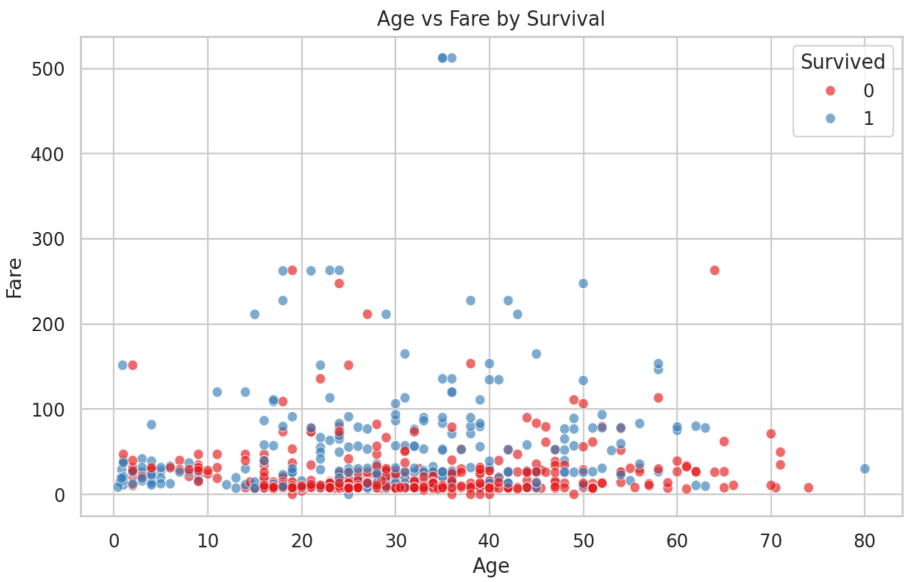


Observation: First-class passengers had the highest survival rate (63.0%), followed by second class (47.3%) and third class (24.2%).

4. Distribution and Relationship Analysis

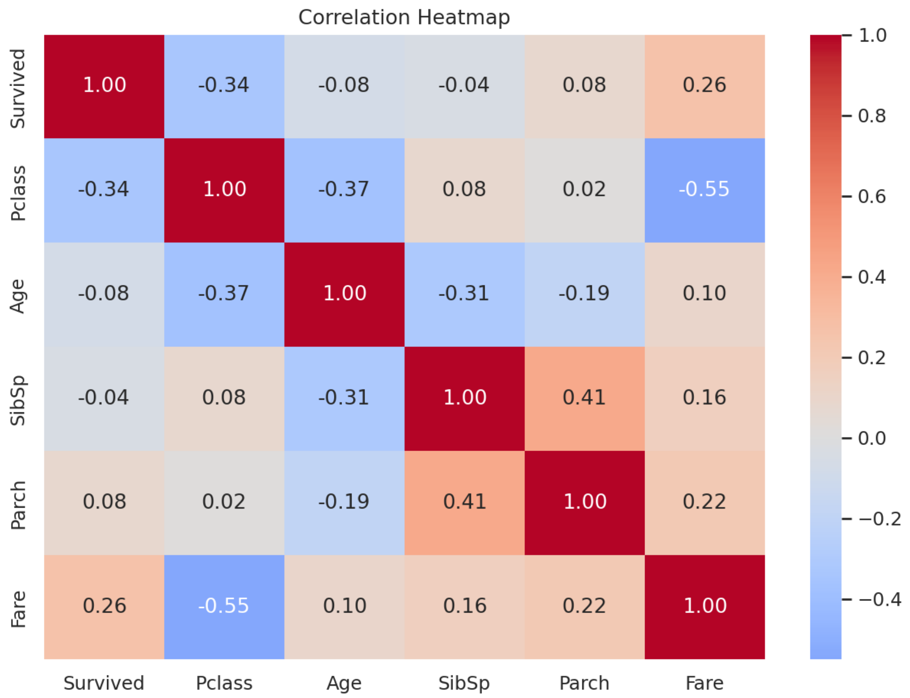


Observation: First-class fares are substantially higher and more variable. High-fare outliers are visible.

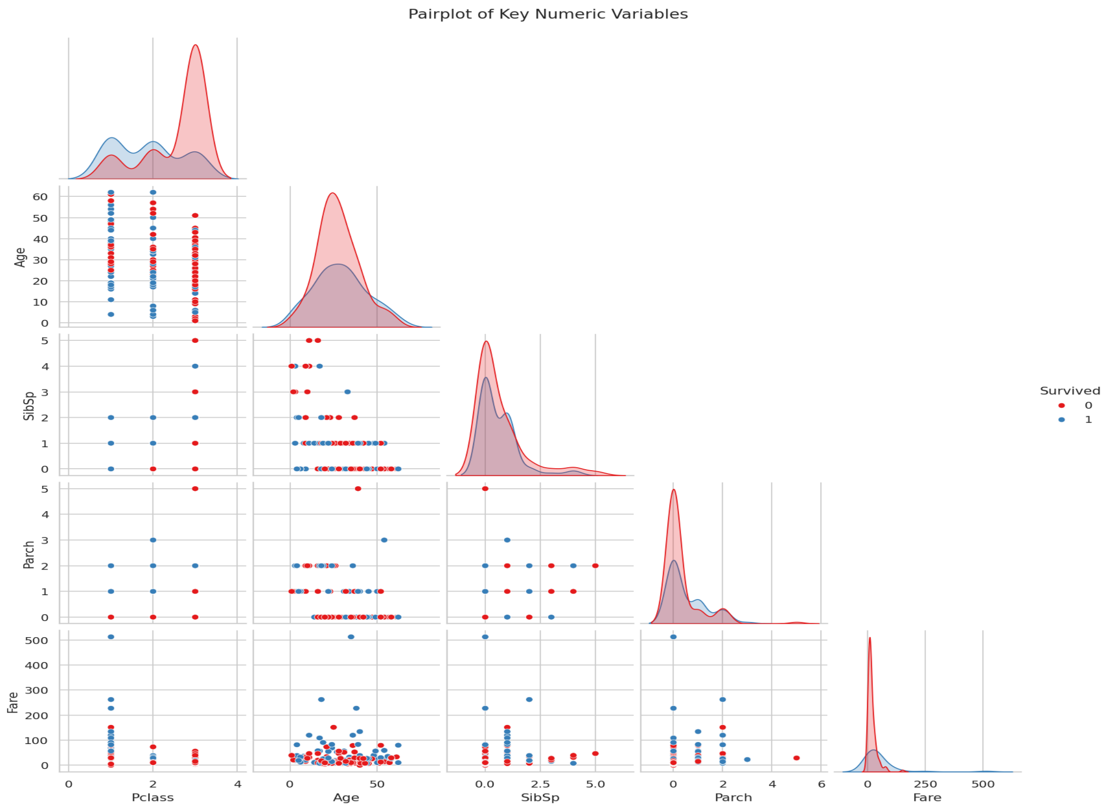


Observation: Age and fare do not show a strong simple linear relationship; survival groups overlap.

5. Correlation and Multivariate Analysis



Key correlations with survival: Pclass = -0.34, Fare = 0.26, Age = -0.08, SibSp = -0.04, and Parch = 0.08. Correlation indicates association, not causation.



The pairplot provides a multivariate view. The clearest survival differences are associated with passenger class and fare, while other numeric variables show considerable overlap.

6. Skewness, Outliers and Data Quality

Fare is highly right-skewed (skewness ≈ 4.79), while SibSp (≈ 3.70) and Parch (≈ 2.75) are also right-skewed. For further modeling, transformations such as log1p may be considered for suitable positive variables. The original dataset was retained unchanged for this EDA.

Multicollinearity should be checked using a correlation matrix and, for formal modeling, Variance Inflation Factor (VIF). The selected numeric predictors do not show extreme pairwise correlations, although Pclass and Fare are meaningfully related.

7. Key Findings

• The dataset has 891 passengers and a 38.38% survival rate.

• Sex is strongly associated with survival; female passengers had a much higher survival rate.

• Passenger class is strongly associated with survival; first class had the highest survival rate.

• Fare is strongly right-skewed and contains high-value outliers.

• Pclass has the strongest negative linear correlation with survival among the selected numeric variables; Fare has the strongest positive correlation.

• Missing values are concentrated in Cabin and Age.

• EDA identifies patterns and associations but does not establish causation.

Conclusion: The Titanic dataset demonstrates a complete EDA workflow using statistical summaries and visual exploration. The most prominent patterns involve sex and passenger class, while fare distribution and missing values are important data-quality considerations.